

**BAGOTVILLE, QC, Canada**

WMO: 717270

Lat: 48.335N

Lon: 70.990W

Elev: 522

StdP: 14.42

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

**Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -20.3      | -15.7 | -27.4                         | 1.2 | -19.2 | -22.7 | 1.6 | -15.2 | 32.1                  | 6.8  | 29.6 | 6.5  | 6.8                   | 270  | 0.671 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.4                   | 84.9            | 67.2 | 81.3 | 65.5 | 78.0 | 64.2 | 70.2                | 79.5 | 68.3 | 77.0 | 66.6 | 74.6 | 10.4                 | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 67.0                            | 101.4 | 74.2 | 65.3 | 95.3 | 72.3 | 63.4 | 89.2 | 70.7 | 34.5          | 79.8 | 32.9 | 77.4 | 31.5 | 74.3 | 80.4              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 27.1              | 24.0 | 21.3 | -26.6                      | 91.4 | 5.5                | 3.3 | -30.5                                              | 93.8 | -33.7      | 95.7 | -36.7      | 97.5 | -40.7      | 99.9 |
|                   |      |      | -26.6                      | 74.4 | 5.1                | 2.3 | -30.3                                              | 76.1 | -33.3      | 77.4 | -36.2      | 78.7 | -40.0      | 80.4 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 38.3   | 5.6   | 9.5   | 21.4  | 36.0  | 50.7  | 61.0  | 66.0  | 63.9  | 55.4  | 42.9  | 29.7  | 15.4  |  |  |
|                                                                        | DBStd       | 22.79  | 13.59 | 12.25 | 11.66 | 8.28  | 7.92  | 7.16  | 5.37  | 5.71  | 7.53  | 7.55  | 9.32  | 11.84 |  |  |
|                                                                        | HDD50       | 5872   | 1377  | 1135  | 887   | 422   | 88    | 5     | 0     | 0     | 29    | 246   | 611   | 1072  |  |  |
|                                                                        | HDD65       | 9956   | 1842  | 1555  | 1352  | 870   | 449   | 161   | 53    | 90    | 301   | 686   | 1060  | 1537  |  |  |
|                                                                        | CDD50       | 1591   | 0     | 0     | 1     | 3     | 109   | 335   | 494   | 431   | 192   | 25    | 1     | 0     |  |  |
|                                                                        | CDD65       | 197    | 0     | 0     | 0     | 0     | 5     | 41    | 82    | 56    | 13    | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 1770   | 0     | 0     | 0     | 2     | 123   | 452   | 651   | 427   | 112   | 3     | 0     | 0     |  |  |
|                                                                        | CDH80       | 435    | 0     | 0     | 0     | 0     | 31    | 127   | 163   | 95    | 19    | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 9.5    | 10.3  | 10.5  | 10.7  | 10.9  | 10.0  | 8.8   | 7.7   | 7.5   | 8.1   | 9.6   | 10.2  | 9.7   |  |  |
|                                                                        | PrecAvg     | 37.7   | 2.4   | 2.1   | 2.2   | 2.3   | 3.1   | 3.6   | 4.6   | 4.0   | 4.0   | 3.3   | 3.0   | 3.0   |  |  |
|                                                                        | PrecMax     | 48.0   | 4.8   | 4.4   | 5.0   | 4.2   | 7.1   | 6.3   | 12.6  | 9.6   | 8.9   | 5.9   | 5.8   | 5.6   |  |  |
| (15) Precipitation                                                     | PrecMin     | 29.7   | 0.7   | 0.5   | 0.5   | 0.5   | 0.6   | 0.7   | 1.5   | 1.4   | 1.6   | 1.1   | 0.9   | 1.1   |  |  |
|                                                                        | PrecStd     | 4.1    | 0.9   | 0.9   | 1.0   | 1.0   | 1.3   | 1.3   | 1.8   | 1.6   | 1.4   | 1.1   | 1.0   | 1.1   |  |  |
|                                                                        | DB          | 40.2   | 40.7  | 51.6  | 66.8  | 84.5  | 89.1  | 88.7  | 87.2  | 82.6  | 71.6  | 58.1  | 41.0  |       |  |  |
|                                                                        | MCWB        | 38.0   | 37.6  | 41.7  | 51.1  | 62.3  | 67.9  | 69.3  | 68.4  | 68.0  | 59.9  | 52.0  | 38.6  |       |  |  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 35.1   | 34.6  | 43.8  | 58.8  | 77.0  | 83.7  | 84.5  | 82.7  | 77.2  | 64.2  | 49.2  | 36.6  |       |  |  |
|                                                                        | 2% MCWB     | 33.9   | 32.4  | 37.1  | 45.2  | 59.2  | 64.8  | 67.9  | 67.5  | 66.1  | 56.1  | 45.4  | 34.6  |       |  |  |
|                                                                        | 5% DB       | 30.3   | 31.9  | 40.0  | 52.6  | 71.0  | 79.1  | 80.7  | 78.6  | 72.2  | 57.9  | 44.8  | 33.9  |       |  |  |
|                                                                        | 5% MCWB     | 29.0   | 30.2  | 34.9  | 42.4  | 55.7  | 63.4  | 66.0  | 65.5  | 62.2  | 51.4  | 41.6  | 32.5  |       |  |  |
|                                                                        | 10% DB      | 25.1   | 28.4  | 36.7  | 47.9  | 65.6  | 75.0  | 77.3  | 75.3  | 68.0  | 53.9  | 41.3  | 31.6  |       |  |  |
|                                                                        | 10% MCWB    | 23.7   | 27.0  | 33.5  | 40.1  | 53.1  | 60.9  | 64.7  | 63.7  | 60.0  | 49.0  | 38.6  | 30.4  |       |  |  |
|                                                                        | 0.4% WB     | 38.5   | 38.7  | 45.4  | 52.6  | 65.9  | 71.8  | 73.5  | 71.9  | 70.5  | 61.5  | 52.6  | 38.9  |       |  |  |
|                                                                        | 0.4% MCDB   | 39.9   | 40.2  | 49.5  | 61.5  | 76.4  | 84.7  | 84.5  | 81.9  | 78.7  | 68.9  | 57.7  | 40.6  |       |  |  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 33.7   | 32.9  | 38.4  | 46.7  | 61.5  | 68.1  | 70.4  | 69.3  | 67.2  | 56.7  | 46.2  | 34.8  |       |  |  |
|                                                                        | 2% MCDB     | 35.1   | 34.4  | 42.6  | 55.7  | 72.1  | 78.4  | 79.3  | 78.5  | 74.7  | 63.5  | 48.8  | 36.3  |       |  |  |
|                                                                        | 5% WB       | 29.2   | 30.2  | 35.7  | 43.6  | 57.6  | 65.7  | 68.4  | 67.5  | 63.8  | 52.6  | 41.9  | 32.7  |       |  |  |
|                                                                        | 5% MCDB     | 30.3   | 31.9  | 38.8  | 50.8  | 68.4  | 74.9  | 77.0  | 75.5  | 70.8  | 56.7  | 44.3  | 33.9  |       |  |  |
|                                                                        | 10% WB      | 23.7   | 26.9  | 33.4  | 41.2  | 54.4  | 63.3  | 66.5  | 65.7  | 61.1  | 49.4  | 38.9  | 30.5  |       |  |  |
|                                                                        | 10% MCDB    | 25.0   | 28.5  | 36.5  | 46.8  | 63.8  | 72.1  | 74.4  | 72.7  | 66.6  | 53.1  | 41.2  | 31.7  |       |  |  |
|                                                                        | MDBR        | 17.0   | 17.5  | 17.3  | 16.8  | 20.5  | 20.0  | 18.4  | 18.5  | 17.8  | 13.5  | 11.6  | 13.4  |       |  |  |
|                                                                        | 5% DB       | 18.9   | 16.5  | 19.5  | 25.8  | 30.2  | 27.4  | 24.0  | 24.1  | 24.5  | 21.4  | 16.4  | 13.4  |       |  |  |
| (36) Mean Daily Temperature Range                                      | 5% DB       | 18.1   | 15.1  | 14.5  | 15.9  | 16.0  | 13.1  | 11.2  | 11.7  | 14.2  | 14.2  | 13.6  | 12.7  |       |  |  |
|                                                                        | 5% WB       | 18.7   | 16.1  | 16.6  | 22.4  | 26.0  | 21.8  | 20.1  | 19.4  | 21.6  | 19.3  | 15.1  | 13.5  |       |  |  |
|                                                                        | MCWBR       | 18.0   | 14.9  | 13.0  | 14.9  | 15.5  | 12.5  | 11.1  | 10.7  | 14.0  | 13.9  | 13.2  | 12.9  |       |  |  |
|                                                                        | MCWBR       | 18.0   | 14.9  | 13.0  | 14.9  | 15.5  | 12.5  | 11.1  | 10.7  | 14.0  | 13.9  | 13.2  | 12.9  |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.265  | 0.284 | 0.315 | 0.380 | 0.361 | 0.384 | 0.404 | 0.383 | 0.371 | 0.329 | 0.314 | 0.266 |       |  |  |
|                                                                        | taud        | 2.270  | 2.224 | 2.201 | 2.200 | 2.401 | 2.365 | 2.323 | 2.383 | 2.405 | 2.505 | 2.388 | 2.322 |       |  |  |
|                                                                        | Ebn at Noon | 255    | 276   | 284   | 275   | 286   | 278   | 271   | 272   | 264   | 260   | 235   | 240   |       |  |  |
|                                                                        | Edh at Noon | 23     | 31    | 37    | 42    | 36    | 37    | 38    | 35    | 31    | 24    | 21    | 20    |       |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 416    | 721   | 1115  | 1413  | 1633  | 1743  | 1648  | 1403  | 1076  | 617   | 337   | 275   |       |  |  |
|                                                                        | RadStd      | 51     | 95    | 92    | 111   | 158   | 117   | 162   | 93    | 81    | 62    | 35    | 29    |       |  |  |

**Historical Trends**

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (27 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page